

09-09-2026

$$Y \text{ random} \quad E[Y] = \int y f_Y(y) dy$$

$$\text{Var}(Y) = \int (y - E(Y))^2 f_Y(y) dy$$

$$Y = E[Y] + \underbrace{Y - E[Y]}_e$$

with information X ,

$$E[Y|X] = \int y f_{Y|X}(y) dy$$

$$\text{Var}(Y|X) = E[(Y - E(Y|X))^2 | X]$$

Law of Iterated Expectation.

$$(i) \quad X = (X_1, X_2)$$

$$E[Y|X_1, X_2] | X_1 = E[Y|X_1]$$

$$(ii) \quad E[Y] = E_X[E(Y|X)]$$

(iii) Conditioning Theorem.

$$E(Y) < \infty, \quad \underbrace{E[g(X) Y | X]} = g(X) E(Y|X)$$

$$E_X[E[g(X) Y | X]] = E[g(X) Y]$$

$$Y = E[Y] + (Y - E[Y])$$

$$= \underbrace{E(Y|X)}_{m(x)} + \underbrace{(Y - E(Y|X))}_e$$

conditional expectation function.

$$E(e|X) = 0$$

mean independence
conditional mean restriction

$$Y = E[Y|X] + E(e)$$

$$E(Y|X) = E[Y|X] + \underbrace{E(e|X)}$$

$$Y \quad E(e|X) = 0$$

$$E(h(X)e) = 0$$

$$E_x[E(h(X)e|X)] \rightarrow$$

$$E_x[h(X) \underbrace{E(e|X)}_{=0}] = 0$$

$$E[Y | X = x]$$

$$\text{Var}(Y | X = x)$$

$$E(e^2) = E_x(E(e^2(X) | X)) = E_x(\sigma^2(X))$$

$m(x) = E(Y|x)$ is the best predictor given information x .

For any other $g(x)$

$$E[(Y - g(x))^2] \geq E(e^2)$$

$$Y = m(x) + e$$

$$Y - g(x) = m(x) + e - g(x)$$

What is the best linear predictor,

$$m(x) = x' \beta$$

$$S(b) = E[(Y - x' b)^2]$$

$$= E[Y^2] - 2b' E[XY] + b' E[XX'] b$$

β = optimal b solves the FOC

$$\beta = E[XX']^{-1} E[XY]$$

$$m(x) = E(Y|x)$$

$$\approx x \beta \approx x_1 \beta_1 + \dots + x_k \beta_k$$

Chapter 3

Random samples.

(Y_i, X_i) are iid draws from the population

iid : independent, identically distributed

$$m(x) \approx X_1 \beta_1 + X_2 \beta_2 + \dots + X_K \beta_K \\ = X' \beta$$

Linear CEF implies a parametric model

$$Y = X \beta + e$$

$$e = \sigma(X) u$$

$$\text{Var}(u) = 1$$

$\rightarrow$ heteroskedastic.
(unequal var).

Goal : estimate β

Parameter : θ is a function of population distribution F .

eg. β in linear prediction is function of $E[XX']$ and $E[XY']$,

Statistic : function of sample. $X_1, \dots, X_n$

An estimator: $\hat{\theta}$ for θ is a statistical rule for computing (guessing) θ .

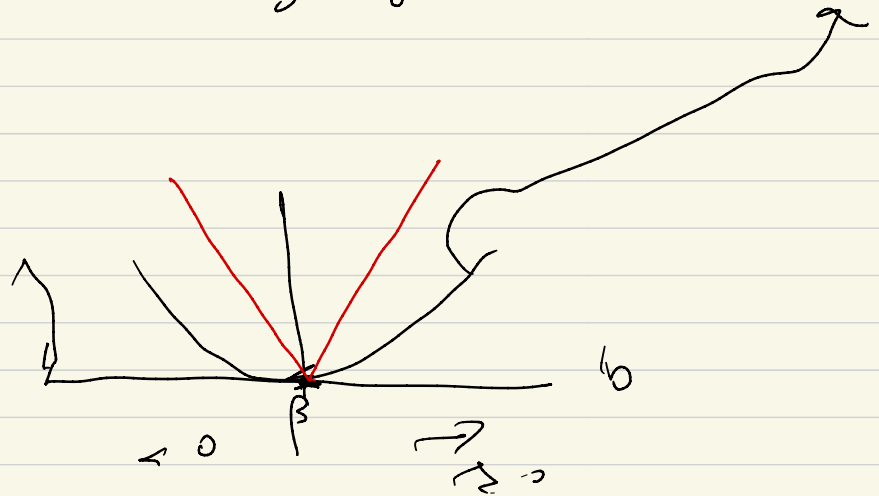
An estimate value when the estimator is calculated from realized data.

Three estimators

① analog principle. / method of moments,

$$X_1 \dots X_n$$
$$E(X) = \mu$$
$$\frac{1}{n} \sum_i X_i = \hat{\mu}$$

② loss function: eg. square loss, $\sum (y_i - x_i b)^2$



③ fully parametric analysis MLE.
eg. assumes $e_i \sim N(0, \sigma^2)$.

Sample regression problem

$$(y_i, x_i) \quad i = 1 \dots n.$$

$$y_i = x_i' \beta + e_i$$

$$= x_{i1} \beta_1 + \dots + x_{ik} \beta_k + e_i.$$

y_i : dependent variable.

x_i : covariates, regressors,
predictors.

Ass 3.1. $\{(y_i, x_i)\}_i$ are iid.

Data generat. process.

$$y = x' \beta + e; \quad E[e_i | x_i] = 0$$

$$E[e_i | x_i] = E[e_i | x_{i1}, \dots, x_{ik}] = 0.$$

$$\text{notation: } \underset{1 \times 1}{y_i} = \underset{1 \times k}{x_i'} \underset{k \times 1}{\beta} + e_i$$

$$\underset{k \times 1}{x_i} = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{ik} \end{bmatrix}$$
$$\underset{k \times 1}{\beta} = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}$$

$$Y = X\beta + e$$

$n \times 1$ $n \times 1$ $K \times 1$ $n \times 1$

$$Y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \quad X = \begin{bmatrix} x_1' \\ x_2' \\ \vdots \\ x_n' \end{bmatrix}$$

Identification.

A parameter can be estimated
only if it is identifiable
(uniquely determined)

$$Y = x_1 \beta_1 + x_2 \beta_2 + x_3 \beta_3 + x_4 \beta_4 + e$$

$$x_4 = x_2 + x_3$$

$$Y = x_1 \beta_1 + x_2 (\beta_2 + \beta_4) + x_3 (\beta_3 + \beta_4) + e$$

